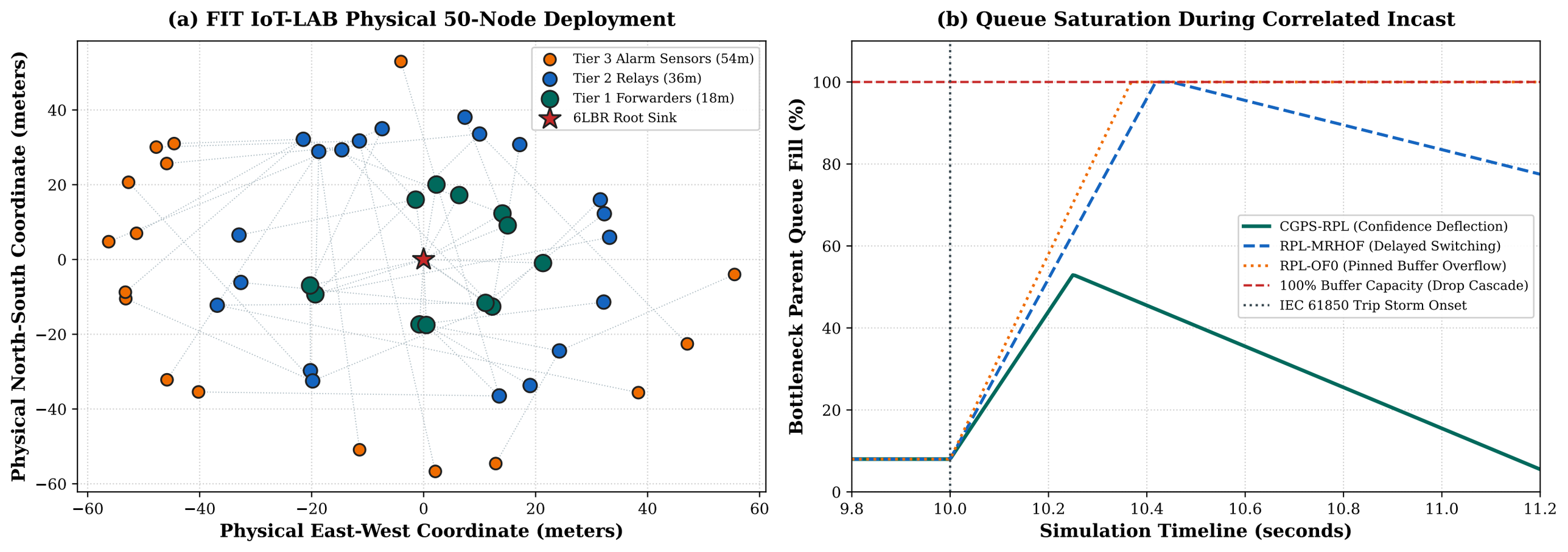
(a) FIT IoT-LAB Physical 50-Node Deployment
Tier 3 Alarm Sensors (54m)
Tier 2 Relays (36m)
Tier 1 Forwarders (18m)
6LBR Root Sink
Physical North-South Coordinate (meters)
Physical East-West Coordinate (meters)
−60
−40
−20
0
20
40
60
(b) Queue Saturation During Correlated Incast
CGPS-RPL (Confidence Deflection)
RPL-MRHOF (Delayed Switching)
RPL-OF0 (Pinned Buffer Overflow)
100% Buffer Capacity (Drop Cascade)
IEC 61850 Trip Storm Onset
Bottleneck Parent Queue Fill (%)
Simulation Timeline (seconds)
0
20
40
60
80
100
9.8
10.0
10.2
10.4
10.6
10.8
11.0
11.2